

MICROBIOLOGY



CHAPTER 22 :

OTHER GRAM NEGATIVE BACTERIA



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Reference:
Dr.Abd Al Kareem
video
and Dr.Sereen old
records

Chapter 22: Other Gram Negative Bacteria

We're going to learn about:

- A. Photosynthetic bacteria
- B. Nitrifying bacteria
- C. Sulfur-oxidizing bacteria

A. Photosynthetic bacteria

Photosynthesis (Carbon fixation): is the process by which the inorganic substances (carbon dioxide and water) are converted to organic compounds with the aid of light energy

Photosynthetic bacteria are divided into two main groups:

- Anoxygenic:

Live in environments that lack oxygen → Two main groups:

1) Purple:

- Purple sulfur bacteria → 2 γ -proteobacterial families: *Chromatiaceae* and *Ectothiorhodospiraceae*
- Purple nonsulfur bacteria → distributed between α - and β -proteobacteria

2) Green

- Green sulfur bacteria → phylum *Chlorobi*
- Green Nonsulfur bacteria → phylum *Chloroflexi*

- Oxygenic:

live in environments that have oxygen → only one group:

Phylum *Cyanobacteria*

Photosynthetic bacteria must have a photosynthetic pigment..why?

Because those pigments collect light and convert the solar energy to chemical energy (ATP)

→ the process of making ATP is called **Phosphorylation**

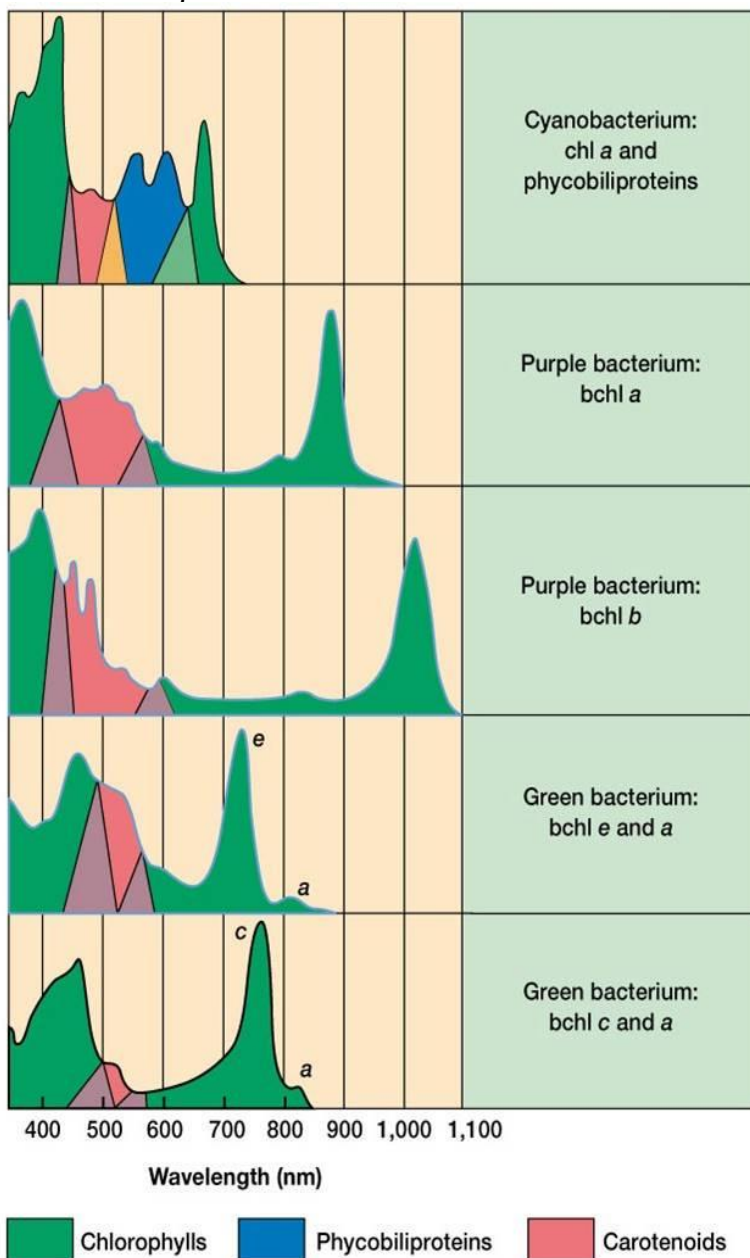
→ BUT when the process of making ATP is with the help of light it's called Photophosphorylation

After studying the chapter..study this table o(•_•)o

Characteristics of the major groups of Gram-negative Photosynthetic Bacteria					
	Anoxygenic Photosynthetic bacteria				Oxygenic photosynthetic bacteria
	Green Sulfur	Green Nonsulfur	Purple Sulfur	Purple nonsulfur	Cyanobacteria
major photosynthetic pigments	Bacteriochlorophylls a plus c, d or e	Bacteriochlorophylls a and c	Bacteriochlorophylls a or b	Bacteriochlorophylls a or b	chlorophyll a plus phycobiliproteins prochlorococcus has divinyl derivatives of chlorophyll a and b
morphology of photosynthetic membranes	photosynthetic system partly in chlorosomes that are independent of the plasma membrane	chlorosomes present when grown anaerobically	photosynthetic system contained in spherical or lamellar membrane complexes that are <u>continuous</u> with the plasma membrane	photosynthetic system contained in spherical or lamellar membrane complexes that are <u>continuous</u> with the plasma membrane	thylakoid membranes lined with phycobilisomes
photosynthetic electron donors	H ₂ , H ₂ S, S	photoheterotrophic donors-a variety sugars, amino acids, and organic acids; photoautotrophic donors: H ₂ S, H ₂	H ₂ , H ₂ S, S	usually organic molecules sometimes reduced sulfur compounds or H ₂	H ₂ O
sulfur deposition	outside of the cell		inside the cell	outside the cell in few cases	inside the cell
nature of photosynthesis	Anoxygenic	Anoxygenic	Anoxygenic	Anoxygenic	oxygenic sometimes facultative anoxygenic
general metabolic type	obligatory anaerobic photolithoautotrophs	usually photoheterotrophic; sometimes photoautotrophic or chemoheterotrophic (when anaerobic and in the dark)	obligatory anaerobic photolithoautotrophs	usually anaerobic photoorganoheterotrophs; some facultative photolithoautotrophs (in the dark, chemoorganoheterotrophs)	aerobic photolithoautotrophs

motility	non-motile; some have gas vacuoles	Gliding	motile with polar flagella; some are peritrichously flagellated	motile with polar flagella or non-motile; some have gas vesicles	non-motile, swimming motility without flagella or gliding motility; some has gas vacuoles
percent G + C	48-58	53-55	45-70	61-72	35-71
phylum or class	<i>Chlorobi</i>	<i>Chloroflexi</i>	α -, β -, and γ -proteobacteria	α -proteobacteria β -proteobacteria (Rhodocyclus)	cyanobacteria

differences in absorption spectra correlates with ecological distribution



This figure shows how different photopigments are present in different groups of photosynthetic microorganisms

If we want to make a growth curve for these microorganisms, we measure the photopigment inside these bacteria with time

يعني منعمل منحنى للوقت مع الامتصاص للضوء من قبل كل صبغة وبما انو كل وحدة عندها نوع صبغات معينة فكل وحدة الها منحنى لحالها

Table 21.2 Procaryotic Bacteriochlorophyll and Chlorophyll Absorption Maxima

This table shows the wave length absorbed by each pigment

*You only have to memorize the second column

Pigment	Long Wavelength Maxima (nm)	
	* In Ether or Acetone	Approximate Range of Values in Cells
Chlorophyll <i>a</i>	665	680–685
Bacteriochlorophyll <i>a</i>	775	850–910 (purple bacteria) ^a
Bacteriochlorophyll <i>b</i>	790	1,020–1,035
Bacteriochlorophyll <i>c</i>	660	745–760
Bacteriochlorophyll <i>d</i>	650	725–745
Bacteriochlorophyll <i>e</i>	647	715–725

^aThe spectrum of bacteriochlorophyll *a* in green bacteria has a different maximum, 805–810 nm.

▪ Anoxygenic:

1) Purple:

- The Purple Nonsulfur Bacteria
 - with one exception (genus *Rhodocyclus*) all are α -proteobacteria
 - morphologically diverse
 - most motile by polar flagella
 - metabolically flexible
 - normally grow anaerobically as anoxygenic photoorganoheterotrophs

Sulfur and nonsulfur means whether the bacteria deposit (ترسب) sulfur or not

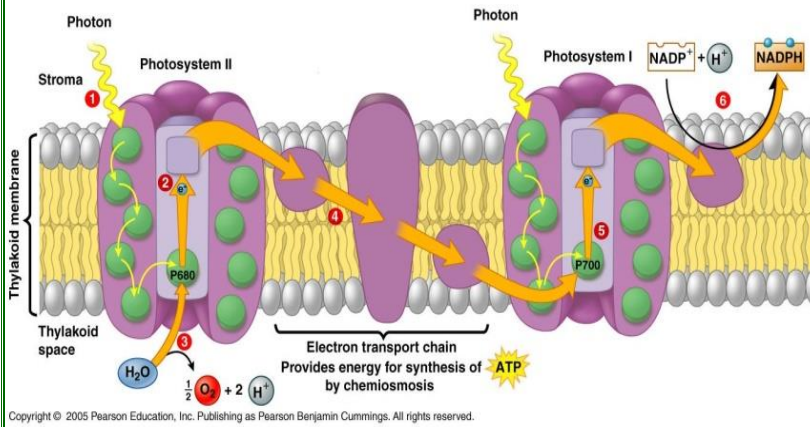
Proteobacteria is a major phylum of Gram-negative bacteria.

بالعادة ينتمو بشكل لا هوائي وبدون وجود الاكسجين.. بهاي الحالة يستخدموا الضوء كمصدر طاقة (photo) والمركبات العضوية كمصدر للالكترونات (organo) ويعملو اختزال (reduction) لمركبات عضوية من كائنات تانية كمصدر للكربون (hetero)

Photoorganoheterotrophs ←

- possess bacteriochlorophylls *a* or *b* in photosystems located in membranes that are continuous with plasma membrane

البناء الضوئي بصير بـ photosystems فيهم الـ bacteriochlorophylls *a* or *b* هذول موجودين ب اغشية (membranes) جاي من الغشاء البلازمي نفسو (يعني بتحمل نفس تركيبو) وهاد التركيب كامل بعمل حويصلة (vesicle) لهيك منحكي انها continuous with the plasma membrane لانها جاي منو



مقتطف اعلاني (افهم ما تحفظ) (w) (

هاي الصورة كلنا درسناها بسنين ماضية، اليروتينات الي لونها بنفسجي هي الphotosystems. اللي عاليسار PhotosystemI واللي عاليمين photosystem II وزى ما هو واضح انو الpigments بكونوا جواهم (الكراات الخضرا)

- some can use low levels of H_2S as electron source

حكينا انها organotroph ولكن بعضهم بستخدموا كميات قليلة من ال H_2S

- in absence of light, most grow aerobically as chemoorganoheterotrophs

لما ما يتوفر الضوء..بتعمل oxidation لمركبات عضوية وغير عضوية عشان تحصل على الطاقة (انتبهوا انها عملت oxidation يعني صارت aerobic)

- in absence of light, some carry out fermentations and grow anaerobically

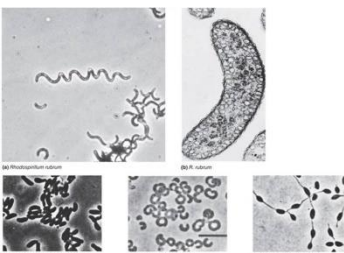
او بتعمل fermentation وبتتموا anaerobically

وهاد كلو دليل على اديه انها ذكية وبتقدر تتكيف مع بيئات مختلفة وتعيش فيها (o~o)

- found in mud and water of lakes and ponds with abundant organic matter and low sulfide levels; some marine species
- Many genera, including Rhodospirillum and Azospirillum, can produce cellular cysts which are:
 - resting cells
 - resistant to desiccation but less tolerant to heat and UV than bacterial endospores
 - made in response to nutrient limitation
 - have thick outer coat and store polyhydroxybutyrate

هدول الcysts يشبهوا الendospores من حيث انهم خاملين(dormant) وبتصنعوا نتيجة لانخفاض في الغذاء ولكن الresistancy تبعتهم اقل من الendospores

→ Some of them are vibrio, some are bacilli and some are spirilla



Purple Sulfur bacteria

- placed in order Chromatiales
 - divided into two families, *Chromatiaceae* and *Ectothiorhodospiraceae*

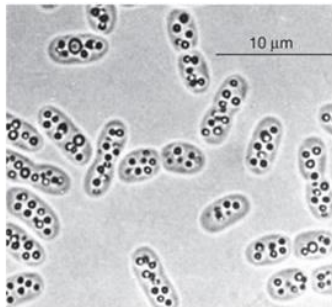
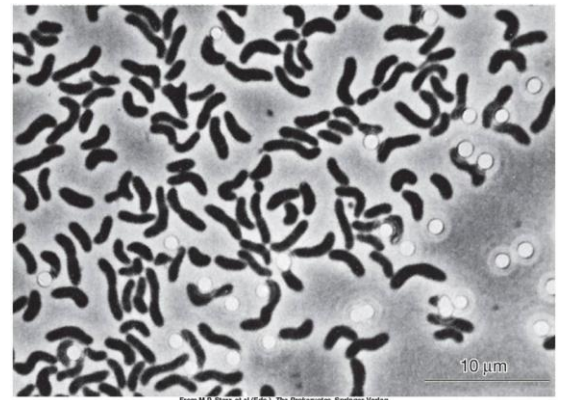
Family *Chromatiaceae*:

- strict anaerobes
- usually photoautolithotrophs
 - use H_2S as electron donor
 - deposit sulfur granules internally
 - often eventually oxidize sulfur to sulfate
 - may also use hydrogen as electron donor
- usually found in anaerobic, sulfide-rich zones of lakes
 - can cause large blooms in bogs and lagoons
- e.g., genera *Thiospirillum*, *Thiocapsa*, and *Chromatium*
 - photosynthetic system contained in spherical or lamellar membrane complexes that are continuous with the plasma membrane

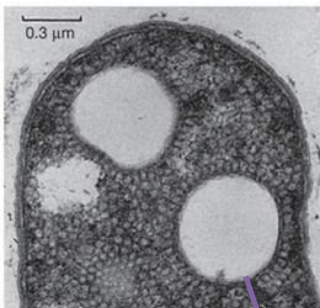
يعني زي الpurple nonsulfur الvesicles الي بتحمل الphotosystems تكون الغشاء تبعها نفس تركيبة الغشاء البلازمي

We find "thio" a lot in their names which means sulfur and it's an indication that they have sulfur granules

a natural view of purple sulfur bacteria. They like H_2S so they live where it can be found → in sewage water.



(a) *Chromatium vinosum*



(b) *C. vinosum*

a: From M.P. Starr, et al (Eds.), *The Prokaryotes*, Springer-Verlag; b: From J.T. Staley, M.P. Bryant, N. Pfennig and J.G. Holt (Eds.), *Bergey's Manual of Systematic Bacteriology*, Vol. 3 Copyright 1996 Williams and Wilkins Co. Baltimore.



sulfur vesicles

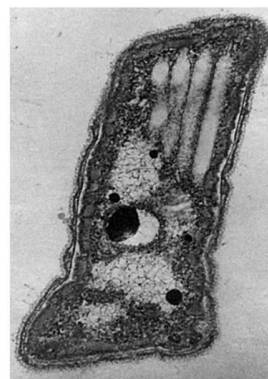
2) Green:

- Green sulfur bacteria (Phylum Chlorobi):

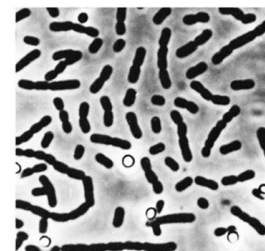
- consists of one class, Chlorobia; one order, Chlorobiales; one family, Chlorobiaceae
- representative genera are Chlorobium, Prosthecochloris, and Pelodictyon
- morphologically diverse
 - rods, cocci, or vibrios; single cells, chains, or clusters
- have chlorosomes and they are
 - ellipsoidal (بيضوي) vesicles attached to plasma membrane
 - bounded by membrane that is not a normal lipid bilayer
 - contain accessory photosynthetic pigments to harvest light energy and convert it to chemical energy
 - **most efficient light harvesting complexes found in nature**

على عكس الpurple ف الgreen ما بتطلع من الغشاء البلازمي ولا نفس تركيبو. الحويصلات هذول بتصنعوا بالخلية بعدين بتصلوا بالغشاء البلازمي وهذول الحويصلات اسمهم Chlorosomes وموجودين بالgreen sulfur والnonsulfur وهمة الي بصير فيهم البناء الضوئي

- lack flagella; nonmotile
 - some have gas vesicles to float over water
 - used to adjust depth of cell for adequate light and H₂S
 - obligately anaerobic
- photolithoautotrophs
- reaction center
- bacteriochlorophyll located in plasma membrane
- use H₂S, elemental sulfur and H₂ as electron sources
 - elemental sulfur deposited outside cell



(a) *Pelodictyon clathratiforme*



(b) *Chlorobium limicola*

- Green Nonsulfur bacteria(Phylum Chloroflexi):

- has both photosynthetic and nonphotosynthetic members
 - e.g., genus Chloroflexus → photosynthetic
 - e.g., genus Herpetosiphon → nonphotosynthetic

→ Genus Chloroflexus:

- filamentous
 - gliding motility
- thermophilic
 - often isolated from neutral to alkaline hot springs
 - grow in form of orange-reddish mats
- metabolism
 - anoxygenic photosynthesis
 - does not use water as electron donor
 - [photoheterotroph](#)
 - can grow aerobically as a [chemoheterotroph](#)

→ Genus Herpetosiphon

- rod-shaped or filamentous
 - gliding motility
- aerobic chemoorganotrophs with respiratory metabolism
- isolated from fresh water and soil habitats

Some additional notes about Anoxygenic bacteria:

1. They only have photosystem II
2. Notice that all of them have the prefix "bacterio" in their pigments names
3. some of them can live aerobically and some anaerobically but all of them do photosynthesis under anaerobic conditions

▪ Oxygenic

Only one Phylum → Cyanobacteria

- largest, most diverse group of photosynthetic bacteria
- Previously called blue green algae
- many are obligate photolithoautotrophs; some can grow slowly in dark as chemoheterotrophs
- one current classification system divides group into 62 species and 24 genera

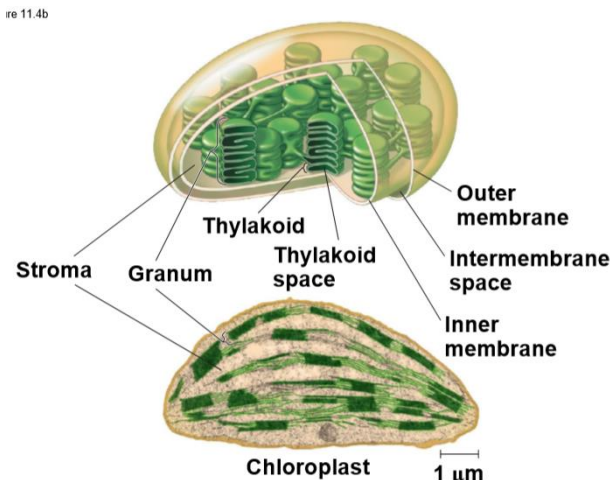
- Photosynthesis in cyanobacteria:
 - resembles that of eucaryotes (the only bacteria that have a chlorophyll similar to what plants and algae have)
 - chlorophyll a
 - prochlorophytes* have chlorophyll a and b
 - have Photosystem II and Photosystem I
 - oxygenic photosynthesis
 - one species can carry out anoxygenic photosynthesis using H_2S as electron source
 - use Calvin cycle to fix CO_2
 - use phycobiliproteins as accessory pigments
 - phycobilisomes, which line thylakoid membranes, contain phycocyanin and phycoerythrin
 - prochlorophytes lack phycobilins

**Prochlorophytes: they lack phycobilisomes as light harvesting structures. Instead, the prochlorophytes have a light-harvesting apparatus composed of the higher plant pigments chlorophylls a and b.*

الcyanobacteria عندها صبغات غير الchlorophyll a اسمهم Phycobiliproteins. Phyco من pigments ويعني صبغات وproteins لانو زي ما منعرف الصبغات هي عبارة عن بروتينات.

هالا Phycobilisomes (somes تعني حويصلات vesicles) بيطنوا الThylakoid membranes من جوا وفيهم الPhycobiliproteins الي منلاقي منهم نوعين بالcyanobacteria: Phycocyanin ولونها ازرق والPhycoerythrin ولونها احمر (Phycoerythrin من Erythrocytes يعني خلايا الدم الحمراء)

مقتطف اعلاني (افهم ما تحفظ) (٧٧٧)



الChloroplasts همة الي بصير فيهم الphotosynthesis

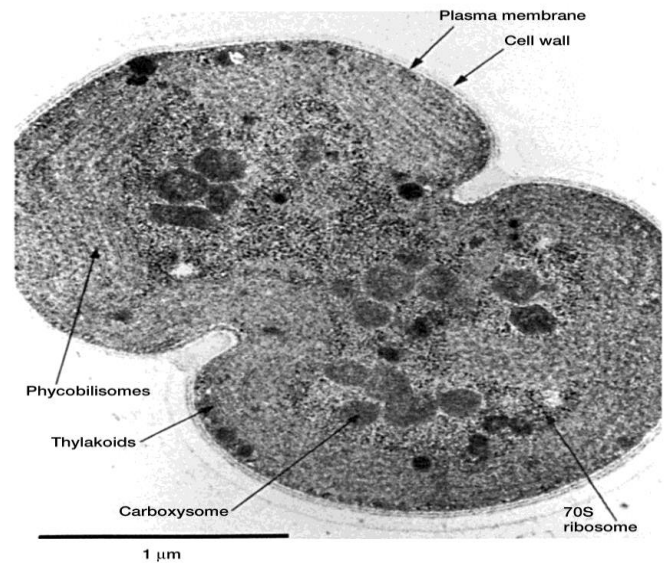
بالنبات وفيهم الThylakoids الي همة الاقراص الي فوق بعض زي ما موضح بالصورة .

بس لازم نتذكر اشي كتير مهم وهو انو احنا منحي عن بكتيريا الي هي prokaryote يعني ما عندها membrane bounded organelles ف الفرق بينهم وبين النبات انو البكتيريا عندها الthylakoid ولكن ما عندها Chloroplasts؛ وبالal thylakoids بصير البناء الضوئي وفيهم منلاقي الchlorophyll a والphycobiliproteins



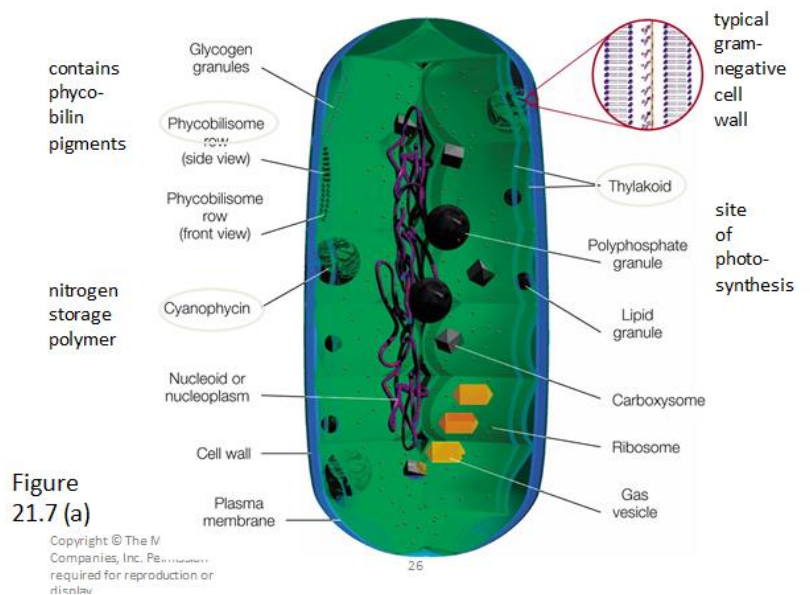
© Elizabeth Gent/Visuals Unlimited

Cyanobacterial Thylakoids and Phycobilisomes



Schematic drawing of a vegetative cyanobacterial cell. The insert shows an enlarged view of the envelope with its outer membrane and peptidoglycan.

Cyanobacteria is photosynthetic and nitrogen fixing bacteria



- range in diameter from ~1 to 10 µm
- may be unicellular, colonial or form filaments called **trichomes** (a row of cells in close contact with each other)
- pigmentation
 - most appear blue-green due to presence of phycocyanin
 - presence of phycoerythrin in many ocean isolates gives them red or brown coloration

- Chromatic Adaptation: modulation of relative amount of pigments
 - phycocyanin production stimulation in the presence of orange light
 - phycoerythrin production stimulation in the presence of blue or blue-green light
- many use gas vacuoles to optimally position themselves in optimal illumination in water column
 - considered a form of phototaxis: movement toward light
- some cyanobacteria are motile by gliding motility and some are not

Different cyanobacteria shapes under the microscope

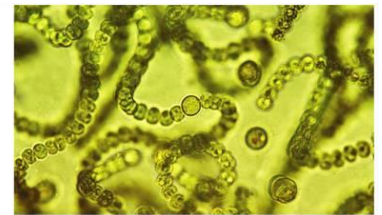
→ Some live in groups

→ Some are filamentous type: filamentous heterocyst and some does not have heterocyst ..just filamentous

→ Some are spirillum



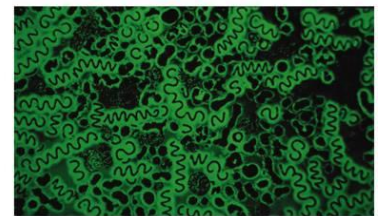
(a) *Chroococcus turgidus*



(b) *Nostoc*



(c) *Oscillatoria*



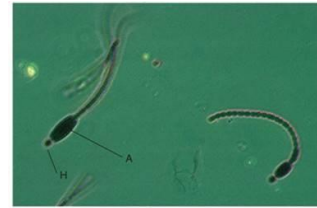
(d) *Anabaena spiroides* and *Microcystis aeruginosa*

a: © T.E. Adams/Visuals Unlimited; b: © Ron Dengler/Visuals Unlimited; c: © M.L. Walker/Photo Researchers, Inc.; d: © T.E. Adams/Visuals Unlimited

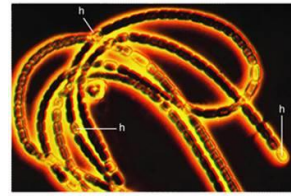
- Specialized reproductive cells and structures:
 - Hormogonia
 - small, motile fragments of filamentous cyanobacteria
 - Akinetes
 - specialized, dormant, thick-walled resting cells that are resistant to desiccation.
 - often germinate to form new filaments
 - baeocytes
 - produced by multiple fission
 - small, spherical cells, which escape when outer wall ruptures
 - some are motile by gliding motility

- Heterocysts:
specialized cells of cayanobacteria used for nitrogen fixation → converting it to ammonia and Enriching the soil as well as their cells by a nitrogen source to produce amino acids and nucleic acids

- produced when organism is nitrogen deprived
- differentiate from individual cells in filament
 - involves reorganization of photosynthetic membranes

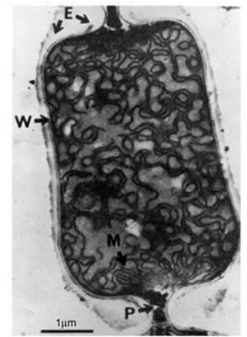


(a) Cylindrospermum



(b) Anabaena

a: © George J. Wilder/Vivante Unlimited; b: Courtesy of Michael Richard, Colorado State Univ.; c: P. Fay and N.J. Lang, Proceedings of the Royal Society London, B178: 185-192, 1971, Norma J. Lang, Univ. of California, Davis



(c) Anabaena heterocyst

Heterocysts and Akinetes

- thick heterocyst wall prevents O_2 diffusion into heterocyst which would inactivate nitrogenase, enzyme responsible for nitrogen fixation (nitrogenase is sensitive to oxygen)

- Classification of Cyanobacteria:
 - unsettled at this time
 - major characteristics include morphology and reproductive patterns
 - other characteristics used
 - ultrastructure
 - genetic
 - physiology and biochemistry
 - habitat/ecology

Table 21.3 Characteristics of the Cyanobacterial Subsections

Subsection	General Shape	Reproduction and Growth	Heterocysts	% G + C	Other Properties	Representative Genera
I	Unicellular rods or cocci; nonfilamentous aggregates	Binary fission, budding	—	31–71	Nonmotile or swim without flagella	<i>Chroococcus</i> <i>Gloeotheca</i> <i>Gleocapsa</i> <i>Prochlorococcus</i> <i>Prochloron</i> <i>Synechococcus</i>
II	Unicellular rods or cocci; may be held together in aggregates	Multiple fission to form baeocytes	—	40–46	Only some baeocytes are motile	<i>Pleurocapsa</i> <i>Dermocarpella</i> <i>Chroococcidiopsis</i>
III	Filamentous, unbranched trichome with only vegetative cells	Binary fission in a single plane, fragmentation	—	34–67	Usually motile	<i>Lyngbya</i> <i>Oscillatoria</i> <i>Prochlorothrix</i> <i>Spirulina</i> <i>Pseudanabaena</i>
IV	Filamentous, unbranched trichome may contain specialized cells	Binary fission in a single plane, fragmentation to form hormogonia	+	38–47	Often motile, may produce akinetes	<i>Anabaena</i> <i>Cylindrospermum</i> <i>Aphanizomenon</i> <i>Nostoc</i> <i>Scytonema</i> <i>Calothrix</i>
V	Filamentous trichomes either with branches or composed of more than one row of cells	Binary fission in more than one plane, hormogonia formed	+	42–44	May produce akinetes; greatest morphological complexity and differentiation in cyanobacteria	<i>Fischerella</i> <i>Stigonema</i> <i>Geitleria</i>

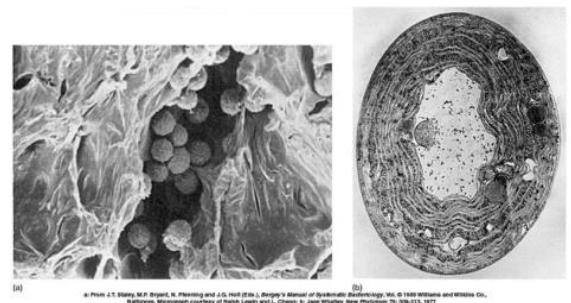
- Prochlorophytes:

→cyanobacteria in genera *Prochloron*, *Prochlorococcus*, and *Prochlorothrix*

- distinguished by presence of chlorophyll a and b and lack of phycobilins
 - are the only prokaryotes to possess chlorophyll b

→makes them candidates as ancestors of endosymbionts that give rise to chloroplasts

Prochloron



- Ecology of cyanobacteria:
 - tolerant of environmental extremes
 - thermophilic species can grow at temperatures up to 75°C
 - often are primary colonizers
 - can cause blooms in nutrient-rich ponds and lakes
 - some produce toxins
 - often form symbiotic relationships
 - e.g., are phototrophic partner in most [lichens](#)
 - e.g., symbionts with protozoa and fungi
 - e.g., nitrogen-fixing species form associations with plants



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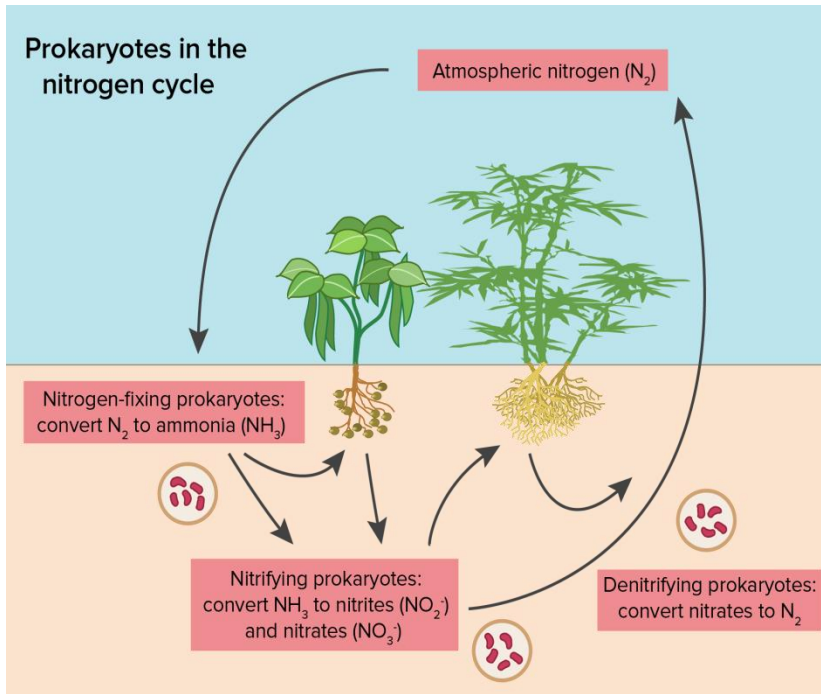
B. Nitrifying Bacteria

- divided into several taxa
 - class *Alphaproteobacteria*; family *Bradyrhizobiaceae* - e.g., genus *Nitrobacter*
 - class *Betaproteobacteria*; family *Nitrosomonadaceae* - e.g., genera *Nitrosomonas* and *Nitrospira*
 - class *Gammaproteobacteria*
 - i. family *Ectothiorhodospiraceae* - e.g., genus *Nitrococcus*
 - ii. family *Chromatiaceae* - e.g., genus *Nitrosococcus*

Table 22.2 Selected Characteristics of Representative Nitrifying Bacteria

Species	Cell Morphology and Size (μm)	Reproduction	Motility	Cytomembranes	G + C Content (mol%)	Habitat
- Ammonia-Oxidizing Bacteria						
<i>Nitrosomonas europaea</i> (β-proteobacteria)	Rod; 0.8–1.1 × 1.0–1.7	Binary fission	—	Peripheral, lamellar	50.6–51.4	Soil, sewage, freshwater, marine
<i>Nitrosococcus oceani</i> (γ-proteobacteria)	Coccoid; 1.8–2.2 in diameter	Binary fission	+; 1 or more subpolar flagella	Centrally located parallel bundle, lamellar	50.5	Obligately marine
<i>Nitrospira briensis</i> (β-proteobacteria)	Spiral; 0.3–0.4 in diameter	Binary fission	+ or —; 1 to 6 peritrichous flagella	Rare	53.8–54.1	Soil
- Nitrite-Oxidizing Bacteria						
<i>Nitrobacter winogradskyi</i> (α-proteobacteria)	Rod, often pear-shaped; 0.5–0.9 × 1.0–2.0	Budding	+ or —; 1 polar flagellum	Polar cap of flattened vesicles in peripheral region of the cell	61.7	Soil, freshwater, marine
<i>Nitrococcus mobilis</i> (γ-proteobacteria)	Coccoid; 1.5–1.8 in diameter	Binary fission	+; 1 or 2 subpolar flagella	Tubular cytomembranes randomly arranged in cytoplasm	61.3 (1 strain)	Marine

From Brenner, D. J.; Krieg, N. R.; and Staley, J. T., Eds. 2005. *Bergey's Manual to Systemic Bacteriology* 2nd ed. Vol. 2: *The Proteobacteria*. Garrity, G. M. Ed-in-Chief. New York: Springer.



مقتطف اعلاني (افهم واحفظ) / (٢٠٠٧)

اول اشئ لازم نمييز الفرق بين Nitrogen fixing و Nitrifying

-**Nitrogen fixing**: converting the atmospheric nitrogen (N_2) to Ammonia (NH_3)

-**Nitrifying**: Converting ammonia to nitrite (NO_2^-) and then to nitrate (NO_3^-)

وهاد هو مايعرف بالNitrogen cycle.

عشان كمية النيتروجين تضل متوازنة. في بكتيريا بتاخذ ال N_2 من الجو وبتحوله لامونيا بالتربة وفي بكتيريا بتاخذ الامونيا وبتحولها ل NO_2^- (يحول بعددين من قبل نوع اخر من البكتيريا ل N_2 ويطلق للجو وما لنا دخل فيهم هدول هلا (● _ ●)).

ال cycle عنا هيك:

$N_2 \rightarrow NH_3 \rightarrow NO_2^- \rightarrow NO_3^- \rightarrow N_2$ من نيتروجين لامونيا ل Nitrite ل Nitrate ل نيتروجين

الجزء الي بهمنا بهاي العملية هو ال Nitrifying bacteria وفي منهم جروبين: (هون بلش احفظ اسمائهم وال Habitats من الجدول)

الاول: Ammonia oxidizing bacteria يعني همة الي بحولو الامونيا ل nitrite وعنا منهم 3:

- 1- Nitrosomonas
- 2- Nitrosococcus
- 3- Nitrospira

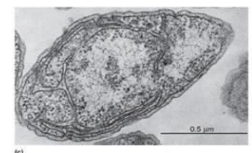
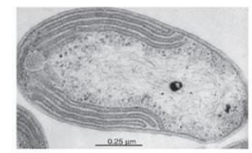
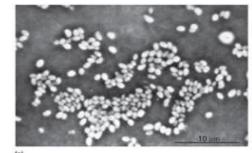
التاني: Nitrite oxidizing bacteria وهمة الي بحولوا ال nitrite ل nitrate وعنا منهم 2:

- 1- Nitrobacter
- 2- Nitrococcus

All of them are Aerobic and catalase Positive

• Nitrification:

- ammonia $\rightarrow$ nitrite $\rightarrow$ nitrate
- conversion of ammonia to nitrate by action of two genera
 - o e.g., Nitrosomonas - ammonia to nitrite
 - o e.g., Nitrobacter - nitrite to nitrate
- fate of nitrate
 - o easily used by plants
 - o lost from soil through leaching or denitrification



(a), (b): © Woods Hole Oceanographic Institution; (c) S.W. Watson, Woods Hole Oceanographic Institution

C. Sulfur-oxidizing bacteria

TABLE 22-5 Sulfur-Oxidizing Genera in Section 21 of *Bergey's Manual*

Genus	Cell Shape	Motility, Flagella	% G + C	Location of Sulfur Deposit ^a	Nutritional Type
<i>Thiobacillus</i>	Rods	+, polar	52–68	Extracellular	Obligate or facultative chemolithotroph
<i>Thiomicrospira</i>	Spirals	+, polar	36–44	Extracellular	Obligate chemolithotroph
<i>Thiobacterium</i>	Rods embedded in gelatinous masses	–	N.A. ^b	Intracellular	Probably chemoorganoheterotroph
<i>Thiospira</i>	Spiral rods, usually with pointed ends	+, polar (single or in tufts)	N.A.	Intracellular	Unknown
<i>Macromonas</i>	Rods, cylindrical or bean shaped	+, polar	67	Intracellular	Probably chemoorganoheterotroph

^aWhen hydrogen sulfide is oxidized to elemental sulfur.

^bN.A., data not available.

- they are NOT photosynthetic and they are colorless. they can deposit sulfur either extracellularly or intracellularly (deposit sulfur in granules inside their cells)
- Some of these microorganisms such as *Thiobacillus* synthesize H₂SO₄ (a very strong acid) and secrete it outside their cells so they are acidophilic microorganisms
 - in addition to those we have the filamentous *Beggiatoa*: filamentous sulfur oxidizing bacteria

The End

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Table 6.2 Sources of Carbon, Energy, and Electrons

Carbon Sources	
Autotrophs	CO ₂ sole or principal biosynthetic carbon source (section 11.3)
Heterotrophs	Reduced, preformed, organic molecules from other organisms (chapters 10 and 11)
Energy Sources	
Phototrophs	Light (section 10.12)
Chemotrophs	Oxidation of organic or inorganic compounds (chapter 10)
Electron Sources	
Lithotrophs	Reduced inorganic molecules (section 10.11)
Organotrophs	Organic molecules (chapters 9 and 10)